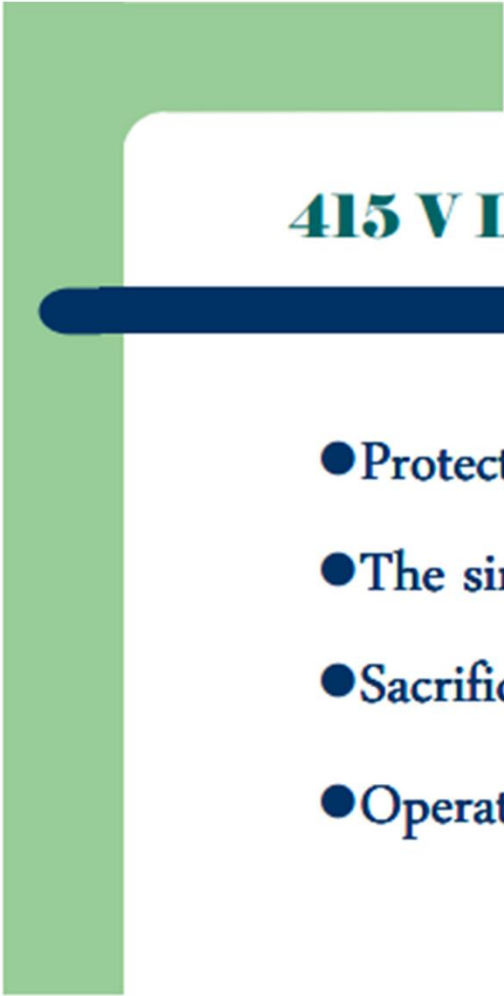


Why Protection?

- Any equipment is subjected to faults both from internal as well as external problems – Component failures, Insulation failures, Lightning strokes, Tree fouling, Bird fouling, Animal electrocution, Abnormal operating conditions, Manual errors, etc.
- Thermal damage
- Need to be protected to avoid damages
- And also to minimize outages from system stability point of view
- Extent of damage depends on both amplitude and duration
- It is essential to know the severity and boundary values
- Voltage and current distribution in a network, due to fault must be evaluated
- Fault calculations will reveal the gravity

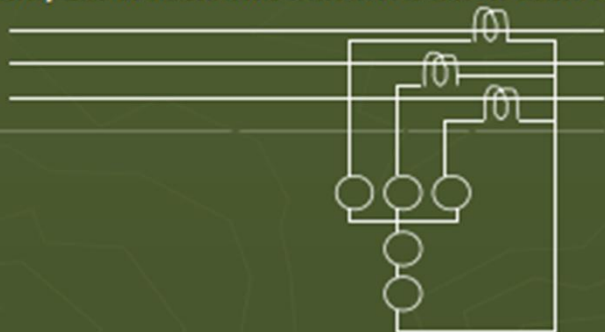


415 V LT Distribution lines

- Protected by Fuses only
- The simplest form of protection
- Sacrificing type
- Operate on actual quantities

HT Distribution lines (11 kV to 33 kV)

- ▶ Actual quantities cannot be handled straightaway – Operate on replica quantities – Instrument transformers needed
- ▶ Protection by employing relays and switchgear
- ▶ Over current, Earth fault and Sensitive earth fault relays are used



DC Scheme used to operate the trip coil

- ▶ O/C relay setting – Ampacity & Line end fault current are the deciding factors
- ▶ E/F relay – Above the expected unbalance
- ▶ SEF relay – Low pick up setting – High time setting – Applicable only for highly balanced lines

WHAT IS SWITCHGEAR?

A device or a combination of devices, primarily intended for the purpose of making, carrying and breaking electric currents in circuits under normal conditions as well as under abnormal (faulty) conditions.

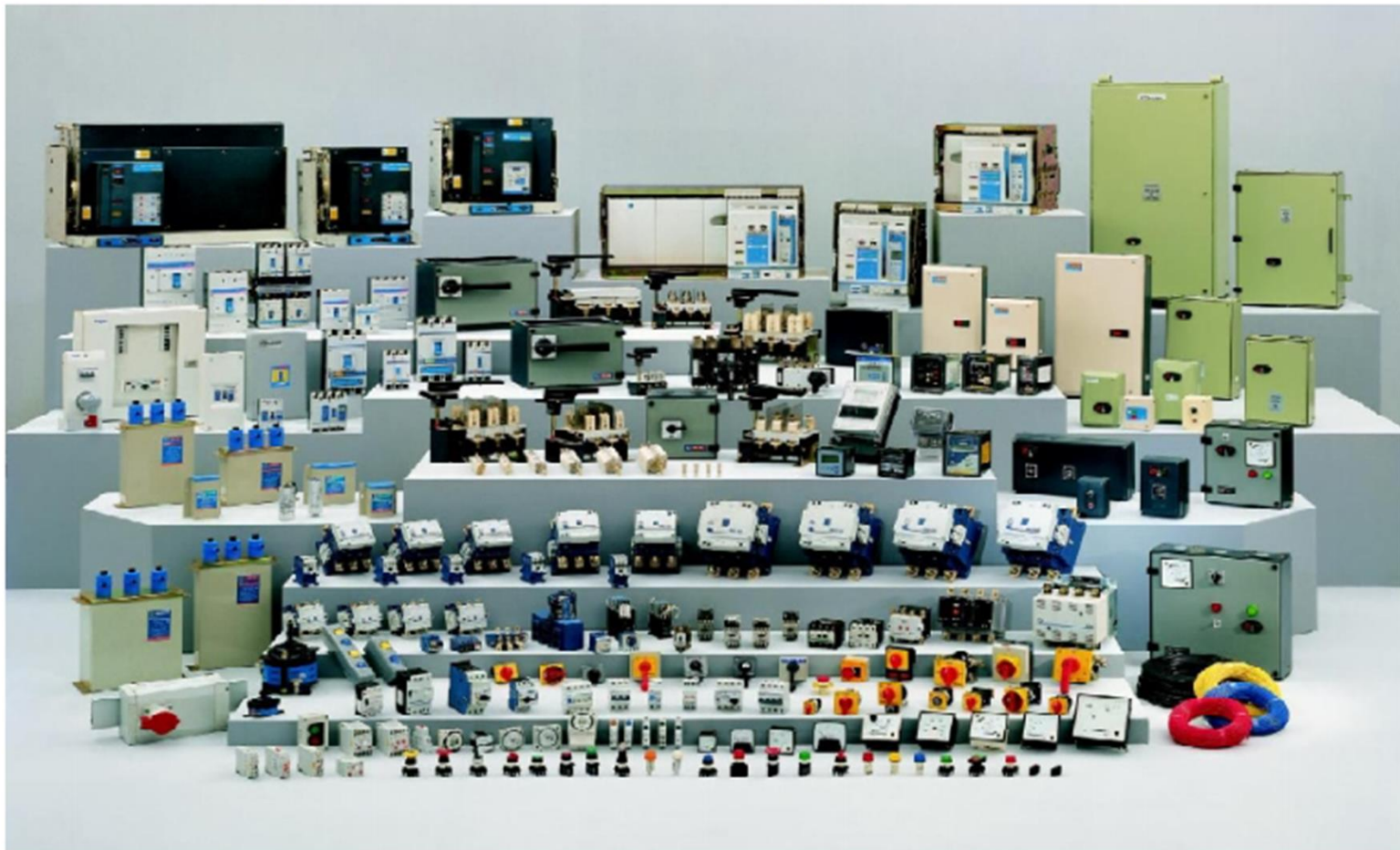
EXAMPLES OF LV SWITCHGEAR

- Air Circuit Breakers (ACBs)
- Moulded Case Circuit Breakers (MCCBs)
- Motor Protection Circuit Breakers (MPCBs)
- Miniature Circuit Breakers (MCBs)
- Earth Leakage Circuit Breakers (ELCBs)
- Residual Current Circuit Breakers (RCCBs)
- Air Break Contactors
- Vacuum Contactors
- Thermal Overload Relays

EXAMPLES OF LV SWITCHGEAR

- Switches/Disconnectors/ Switch-Disconnectors
- HRC Fuses
- Switch Fuse Units (SFU)
- Fuse Switch Units (FSU)
- Switch – Disconnector - Fuse (SDF) Units
- Current Transformers
- Voltage Transformers
- Selector Switches
- Push Buttons

Range of Low Voltage Switchgear



NEED OF PROTECTIVE SYSTEMS

- An EPS consists of Generators, Transformers, Transmission line and Distribution lines, etc..
- SC and other abnormal conditions often occur in Power systems.
- A Protective scheme include CB and Protective relays to isolate the faulty section from the healthy.
- A CB can disconnect the faulty element, when it is called upon to do so by the relay.

RELAY

- It is a device which sense the abnormal conditions on a PS by constantly monitoring electrical quantities.- voltage, current, phase angle(direction) , frequency
- The function of the relay is to detect and locate a fault and issue a command to the CB to disconnect the faulty element.
- A Protective relay does not anticipate or prevent the occurrence of a fault, rather it takes action only after fault has occurred.
- However – Buchholz relay is an exception.

OTHER ABNORMAL CONDITION

- Over speed of generators and motors
 - Over voltage
 - Under frequency
 - Loss of excitation
 - Overheating of excitation of stator and rotor
-
- Note: Cost of protective scheme should be about 5% of the total cost of the systems.

NATURE AND CAUSES OF FAULTS

- Insulation failure - SC
- Conducting path failure
- Over voltage due to lightning/ switching surge
- External conducting objects falling on overhead lines
- Fine dust, cement accumulation – on pin insulator – reduces insulation strength – causes flash over
- Birds body touches line and earth wire
- Joint failure in cable and overhead line
- Unbalanced current – rotating machine – set up harmonic current – heating

NATURE AND CAUSES OF FAULTS

- Other causes are (overhead lines): direct lighting strokes, aircraft, snakes, ice & snow loading, storms, earth quakes
- Other causes(Gen, Transformer): ageing, accidental operation, wrong connection, defect in the protective devices, sometime poor quality of system components, faulty system design .

TYPES OF FAULT

- Symmetrical fault : 3 phase fault (with / without the Ground)- to determine the system fault level
- Unsymmetrical fault:
 - LG – due to failure of insulation, due to phase conductor breaking and falling to the ground
 - L-L-G –
 - L – L
- Open Circuited faults
- Winding fault

Simultaneous fault

- Two or more faults occurring simultaneously
Same/ different type – at same or different point

cross – country earth fault

- The simultaneous presence of an L-G fault at one point and a second L-G fault on another phase at some other point is an example of two faults of the same type at two different points. If these two L-G faults are on the same section of the line, they are treated as L-L-G fault. If they occur in different line sections, it is known as a cross – country earth fault.

Cross-country faults

- Cross-country faults are common on systems grounded through high impedance or Peterson coil but they are rare in a solidly grounded systems.

BASIC RELAY TERMINOLOGY

Relay: A relay is an automatic device by means of which an electrical circuit is indirectly controlled (opened and closed) and is governed by a change in the same or another electrical circuit.

Protective Relay

A Protecting relay is an automatic device which detects an abnormal condition in an electrical circuit and causes a circuit breaker to isolate the faulty element of the systems. In some cases it may give an alarm or visible indication to alert operator

Operating Force or torque

A force or torque which tends to close the contacts of the relay

Restraining force or torque

A force or torque which opposes the
operating force/torque

Actuating quantity

An electrical quantity (current, voltage, etc.) to which a relay responds

Pick-up(level)

The threshold value of the actuating quantity (current , voltage..) above which the relay operates

Reset on drop-out

The threshold value of the actuating quantity below which the relay is de-energised and returns to normal position or state

Operating time

It is the time which elapses from the instant at which the actuating quantity exceeds the relays pick-up value to the instant at which the relay closes its contacts.

Reset time

It is the time which elapses from the moment the actuating quantity falls below its reset value to the instant when the relay comes back to its normal position

Setting

The value of the actuating quantity at which the relay is set to operate

Primary protection

If a fault occurs, it is the duty of the primary protective scheme to clear the fault. It acts as a first line of defence. If it fails, the back-up protection clears the fault

Back up protection

The back-up protection is designed to clear the fault if the primary protection fails. It acts as a second line of defence.

Back-up relay

A back-up relay operates after a slight delay, if the main relay fails to operate.

Instantaneous relay

An instantaneous relay has no intentional time delay in its operation.

Inverse time relay

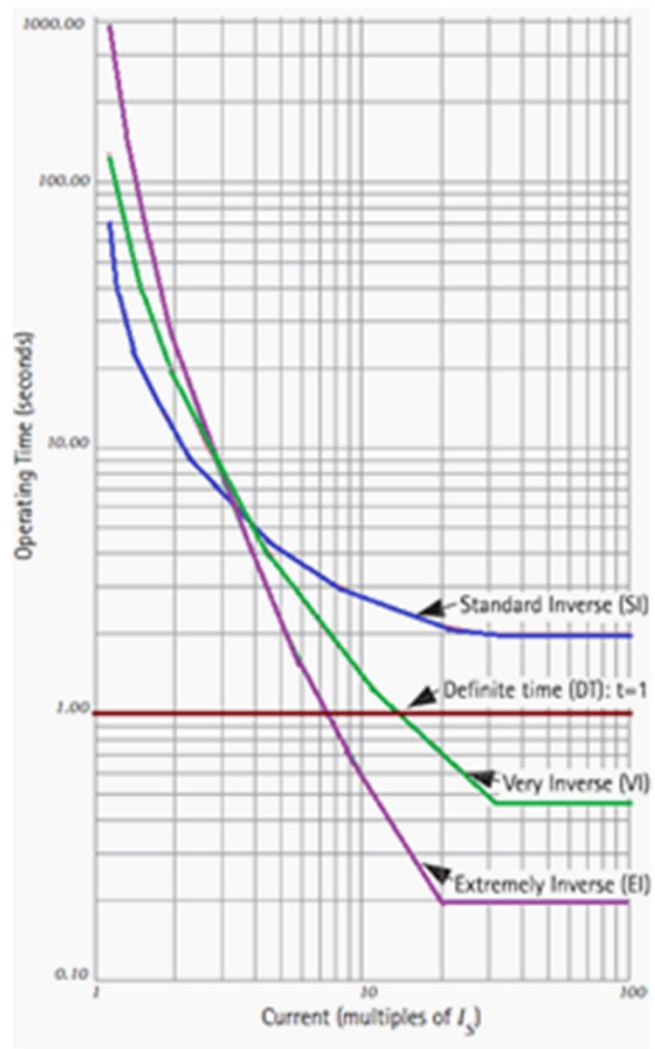
A relay in which the operating time is inversely proportional to the magnitude of the operating current.

Definite time relay

The relay in which the operating time is independent of the magnitude of the actuating current.

Inverse definite minimum time (IDMT) relay

A relay which gives an inverse time characteristics at lower values of the operating current and definite time characteristics at higher values of the operating current.



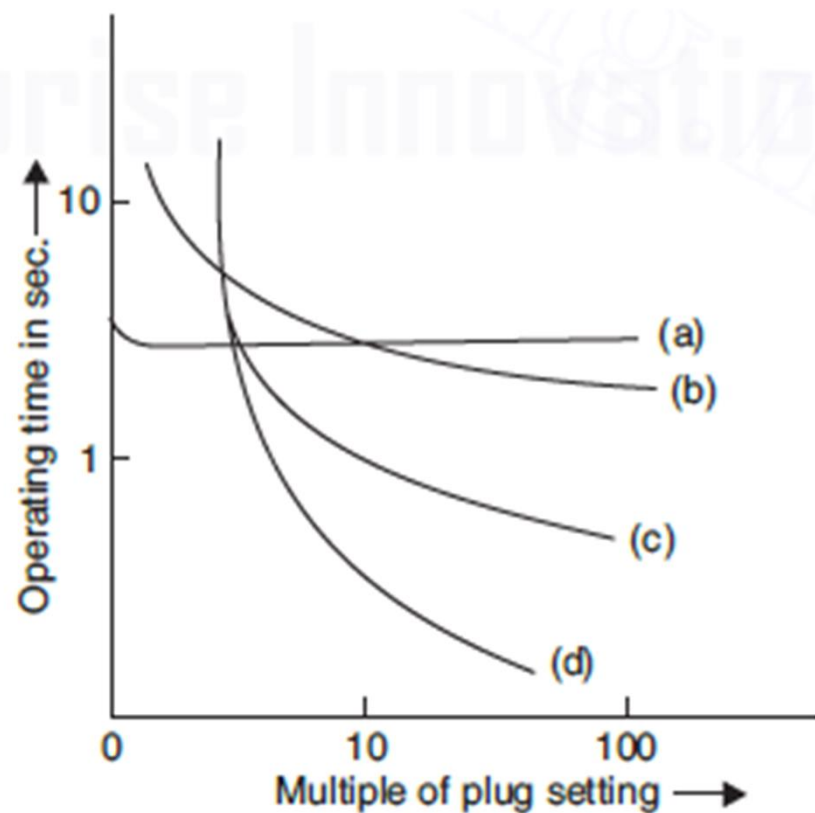


Fig. 14.6 Characteristics of various over-current relays: (a) definite time, (b) IDMT, (c) very inverse, and (d) extremely inverse.

Earth fault relay

A relay used for the protection of an element of a power systems against earth faults is known as an earth fault relay.

Phase fault relay

A relay used for the protection of an element of a power systems against phase faults is known as an phase fault relay.

Negative sequence relay

A relay for which the actuating quantity is the negative sequence current. When the negative sequence current exceeds a certain value, the relay operates. This type of relay is used to protect electrical machines against overheating due to unbalanced current.

Zero sequence relay

A relay for which the actuating quantity is the zero sequence current. This type of relay is used for earth fault protection.

Reach

This term is mostly used in connection with distance relays. A distance relay operates when the impedance as seen by the relay is less than a preset value. This preset impedance or corresponding distance is called the reach of the relay. In other words, it is the maximum length of the line up to which the relay can protect

Over reach

Some time a relay may operate even when a fault point is beyond its present reach (i.e. its protected length)

Under reach

Sometimes a relay may fail to operate even when the fault point is within its reach, but it is at far end of the protected line. This phenomena is called under reach.

Restricted earth fault protection

This is an English term which may be misunderstood in other countries. It is used in the context of transformer or alternator. It refers to the differential protection of transformers or alternators against ground faults. It is called restricted because its zone of protection is restricted only to the winding of the alternator or transformer. The scheme responds to the faults occurring within its zone of protection. It does not respond to faults beyond its zone of protection.

Unrestricted protection

A protection system which has no clearly defined zone of operation and which achieves selective operation only by time grading.

Percentage distribution of Faults in Power Systems

- Overhead line : 50%
- Underground cables:9%
- Transformer: 10%
- Generators:7%
- Switchgears:12%
- CTs, PTs, other control equipment: 12%

Frequency of occurrence of faults

- L-G : 85%
- L-L :8%
- 2L-G:5%
- 3 phase :2%

Protection classification

Protective relays are broadly classified into the following three categories depending on the technologies they use for their construction and operation.

- (i) Electromechanical relays
- (ii) Static relays
- (iii) Numerical relays

There are various types of protective relays in each category, depending on the operating principle and application.

2.2 ELECTROMECHANICAL RELAYS

Electromechanical relays operate by mechanical forces generated on moving parts due to electromagnetic or electrothermic forces created by the input quantities. The mechanical force results in physical movement of the moving part which closes the contacts of the relay for its operation. The operation of the contact arrangement is used for relaying the operated condition to the desired circuit in order to achieve the required function. Since the mechanical force is generated due to the flow of an electric current, the term 'electromechanical relay' is used.

Most electromechanical relays use either electromagnetic attraction or electromagnetic induction principle for their operation. Such relays are called *electromagnetic relays*. Depending on the principle of operation, the electromagnetic relays are of two types, i.e., (i) attracted armature relays, and (ii) induction relays. Some electromechanical relays also use electrothermic principle for their operation and are based upon the forces created by expansion of metals caused by temperature rise due to flow of current. Such relays are called *thermal relays*. Most of the present day electromechanical relays are of either induction disc type or induction cup type.

The following are the principal types of electromechanical relays:

1. Electromagnetic relays
 - (i) Attracted armature relays, and
 - (ii) Induction relays
2. Thermal relays

2.3 STATIC RELAYS

In a static relay, the comparison or measurement of electrical quantities is performed by a static circuit which gives an output signal for the tripping of a circuit breaker. Most of the present day static relays include a dc polarised relay as a slave relay. The slave relay is an output device and does not perform the function of comparison or measurement. It simply closes contacts. It is used because of its low cost. In a fully static relay, a thyristor is used in place of the electromagnetic slave relay. The electromechanical relay used as a slave relay provides a number of output contacts at low cost. Electromagnetic multicontact tripping arrangements are much simpler than an equivalent group of thyristor circuits.

A static relay (or solid state relay) employs semiconductor diodes, transistors, zener diodes, thyristors, logic gates, etc. as its components. Now-a-days, integrated circuits are being used in place of transistors. They are more reliable and compact.

Earlier, induction cup units were widely used for distance and directional relays. Later these were replaced by rectifier bridge type static relays which employed dc polarised relays as slave relays. Where overcurrent relays are needed, induction disc relays are in universal use throughout the world. But ultimately static relays will supersede all electromagnetic relays, except the attracted armature relays and dc polarised relays as these relays can control many circuit at low costs.

2.3.1 Merits and Demerits of Static Relays

The advantages of static relays over electromechanical relays are as follows.

- (i) Low burden on CTs and VTs. The static relays consume less power and in most of the cases they draw power from the auxiliary dc supply
- (ii) Fast response
- (iii) Long life
- (iv) High resistance to shock and vibration
- (v) Less maintenance due to the absence of moving parts and bearings
- (vi) Frequent operations cause no deterioration
- (vii) Quick resetting and absence of overshoot
- (viii) Compact size
- (ix) Greater sensitivity as amplification can be provided easily
- (x) Complex relaying characteristics can easily be obtained
- (xi) Logic circuits can be used for complex protective schemes

The demerits of static relays are as follows:

- (i) Static relays are temperature sensitive. Their characteristics may vary with the variation of temperature. Temperature compensation can be made by using thermistors and by using digital techniques for measurements, etc.
- (ii) Static relays are sensitive to voltage transients. The semiconductor components may get damaged due to voltage spikes. Filters and shielding can be used for their protection against voltage spikes.
- (iii) Static relays need an auxiliary power supply. This can however be easily supplied by a battery or a stabilized power supply.

2.5 COMPARISON BETWEEN ELECTROMECHANICAL RELAYS AND NUMERICAL RELAYS

Sr. No.	Feature	Electromechanical Relay	Numerical Relay
1.	Size	Bigger	Compact
2.	Characteristics	Fixed	Selectable
3.	Flexibility	No flexibility	Flexibility due to programmability.
4.	Communication feature	Not available	Available
5.	Blocking feature	Not available	Available
6.	Self-supervision	Not available	Available
7.	Adaptability	Not adaptable	Adaptable to changing system condition.
8.	Multiple-functions	Not possible	Possible
9.	Accuracy	$\pm 5\%$ or more	$\pm 2\%$
10.	Speed of operation	Slow	Fast
11.	Burden on Transducers (CTs and VTs)	Very high	Extremely low
12.	Consistency of calibration	Deteriorate with time	No effect on calibration even after use of 20-25 years
13.	Setting	Through plug setting in fix steps.	Software based settings.
14.	Memory feature	No memory of any type is available.	Several memory features are available.
15.	Maintenance	Cumbersome and frequent maintenance required.	Maintenance free relays
16.	Output relay programming	Not available	Available
17.	Accessibility of relay from remote place	Not possible	Remote accessibility is available
18.	Status of service values	Not available	Available
19.	Safety of personnel	Not adequate due to non-accessibility at remote location.	Adequate safety is provided.
20.	Spares requirement	There is need to stock several items as spare.	Universal designs minimise the spares requirement.
21.	Upward connectivity for a present or future system such as SCADA, EMS etc.	Not possible	Possible